

MATH 1400 — Quiz 1

Section: _____

Name: _____

Date: _____

Instructions: Show clear work. Final answers should be simplified.

1. Let

$$f(x) = \frac{1}{x+2}, \quad g(x) = \frac{x}{x-1}.$$

Compute $(fg)(x)$.

2. Find the domain of

$$h(x) = \frac{3x}{x^2 - 4}.$$

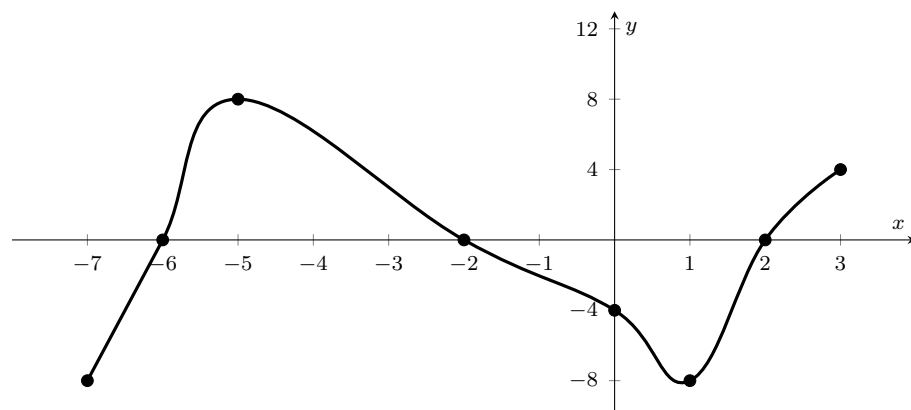
3. Use the Vertical Line Test to explain why

$$x^2 + y^2 = 1$$

does *not* define y as a function of x .

(If you are not comfortable with the Vertical Line Test yet, you may instead give a specific x -value that produces two different y -values.)

4. Let f be the function whose graph is shown below. Answer parts (a) and (b).



(a) What is the range of f ?

(b) For what values of x does $f(x) = -8$?
